Keno: generative noise subtraction for template-free gravitational-wave burst search

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July 23, 2026

Abstract

Keno is a research prototype for detecting short gravitational-wave bursts without assuming a known waveform shape. A U-Net predicts instrumental noise from whitened strain; after subtraction, the residual is searched with an STFT excess-power statistic, a Hanford–Livingston lag scan, and an envelope-timing veto.

The implementation is end to end: PyTorch/FastAPI inference, a Node.js orchestration layer with in-memory caching, and an Angular interface for synchronized inspection of raw strain, predicted noise, and residuals.

On synthetic bursts injected into real LIGO background at 1% false-alarm rate (FAR), Keno residual excess power reaches near 100% detection efficiency from SNR 2 to 12. A deliberately mismatched matched-filter template remains near ∼20% efficiency; a BBH-trained ResNet control evaluated off-template plateaus below saturation. At this FAR, raw excess power without subtraction also saturates, so the injection campaign alone does not separate Keno from raw excess power. On published catalog GPS times, the residual path can trigger where raw excess power does not (e.g. GW150914), and a coherent dual-detector residual search recovers 9 of 18 dual-detector events; GW170817 is rejected by the envelope veto owing to a known Livingston glitch.

Keno is intended to complement morphology-specific binary black hole classifiers such as AresGW, not to replace them. The release bundles model, API, and inspection tooling for reproducible unmodeled residual search.

1 Introduction

LIGO and Virgo record *strain*: a dimensionless measure of spacetime stretch induced by passing gravitational waves. In analysis pipelines this appears as a high-rate one-dimensional time series; here the sampling rate is 4096 Hz. Data are typically *whitened* first—filtered so that stationary background noise has approximately unit variance across frequency, which improves sensitivity to short transients.

Search methods fall into two complementary families:

- **Template-based / morphology-specific.** When the expected signal shape is known (for example a binary black hole merger chirp), one may match-filter against templates or train a classifier on that family. AresGW [Nousi et al., 2023, Koloniari et al., 2025] exemplifies the classifier approach for mergers.
- **Template-free / unmodeled.** When the morphology is unknown (short ringdowns, noise-like bursts, or other transients), the search targets localized excess energy consistent across detectors, without assuming a particular waveform.

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Keno belongs to the second family. A neural network estimates instrumental noise $\hat{N}$ and forms the residual

$$R = S_{\mathrm{raw}} - \hat{N}. \tag{1}$$

Throughout this paper, R denotes that residual (sometimes referred to informally as "cleaned strain"). Detection proceeds via excess power—peak STFT energy relative to a robust noise floor—followed by an H1+L1 lag scan and an envelope-consistency veto to reject loud single-detector glitches.

Keno was developed as an independent research prototype and is not a LIGO collaboration product. It should not be interpreted as a production search pipeline. The repository includes model weights together with ingestion, inference, evaluation scripts, and a browser-based viewer for residual inspection.

Relation to morphology-specific classifiers. Morphology-specific systems such as AresGW [Nousi et al., 2023] test whether a segment resembles a black-hole merger seen in training. Keno tests whether excess energy remains after subtracting the estimated noise component. These address distinct detection problems; Keno is positioned as a complementary tool for unmodeled bursts.

Scope of claims. In this prototype, generative subtraction followed by residual excess-power search recovers unknown-morphology injections at controlled FAR, strongly outperforming a mismatched matched filter and a BBH-trained ResNet control evaluated off-template. Relative to raw excess power, the injection campaign at 1% FAR saturates for both methods; the residual advantage is demonstrated on published catalog GPS times (residual-only single-detector triggers and production dual-detector coincidence).

2 Terminology

The following definitions are provided for readers less familiar with gravitational-wave search terminology.

SNR Signal-to-noise ratio. Higher values correspond to louder injections relative to background.

FAR False-alarm rate: the fraction of pure-noise trials that exceed the detection threshold. At 1% FAR, approximately one noise-only window in one hundred yields a false trigger.

Matched filter Cross-correlation of the data with a known waveform template. Effective when the template matches the signal; weak otherwise.

Excess power A template-free energy statistic: the loudest time–frequency cell in an STFT, normalized by a robust noise floor (here, median STFT power).

Residual R $S_{\mathrm{raw}} - \hat{N}$. Background energy should decrease; a genuine burst should largely survive.

Glitch A non-astrophysical detector artifact. Gravity Spy catalogs common morphological classes (Blip, Koi Fish, Scattered Light, and others).

H1 / L1 LIGO Hanford and Livingston. Requiring both detectors reduces single-detector false alarms.

GWOSC Gravitational Wave Open Science Center, the public HTTP archive from which Keno retrieves open strain frames.

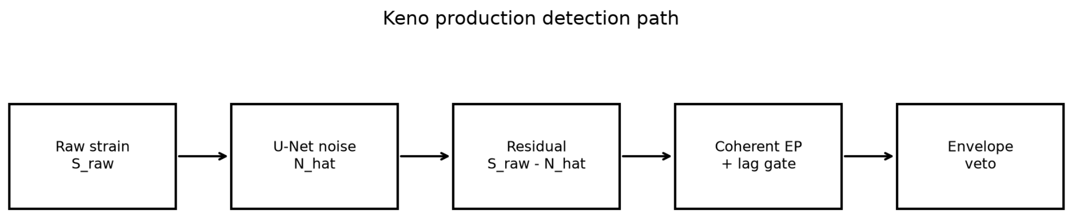

Figure 1: Keno detection pipeline: predict noise $\hat{N}$, form residual $R = S_{\mathrm{raw}} - \hat{N}$, compute residual excess power, and apply H1+L1 timing consistency with an envelope veto.

3 Method

Figure 1 summarizes the detection pipeline. Strain segments are whitened and sampled at $f_s = 4096\,\mathrm{Hz}$. Unless stated otherwise, analysis windows are $T = 4\,\mathrm{s}$ ($N = 16\,384$ samples).

3.1 Software stack

Keno is implemented as three coordinated services.

- **ML engine (Python / FastAPI / PyTorch).** Loads the U-Net checkpoint, fetches and whitens strain via `gwpy`, performs subtraction, and exposes REST endpoints for denoising (`POST /api/v1/denoise`), single-detector detection (`POST /api/v1/detect`), and dual-detector coincidence (`POST /api/v1/detect/coincidence`). Offline evaluation modules (`app.prove`, injection campaigns, glitch stress tests, catalog follow-up) invoke the same residual-search logic as the live API.
- **Backend orchestrator (Node.js / Express).** Proxies the Angular frontend to the ML engine, forwards GWOSC and offline error details, and caches denoise/detect responses in memory. Initial GWOSC downloads may require 1–3 minutes; inference requests therefore use a timeout of 300 s, whereas health checks use 5 s. Repeated queries for the same GPS window typically return from cache within seconds.
- **Frontend (Angular).** Displays raw strain, predicted noise $\hat{N}$, and residual R on synchronized axes for preset catalog events (e.g. GW150914) or user-specified GPS times. The interface also reports blind-search quantities: raw versus residual excess power, fraction of threshold, and H1+L1 coincidence flags. A synthetic validation mode injects a known burst to verify the subtraction path without a live download.

GWOSC ingestion is centralized in the ML service: frames surrounding a GPS timestamp are fetched, validated for NaN gaps on offline detectors, whitened, and returned as arrays. Cached background segments used for training and FAR calibration are stored on disk so that evaluation campaigns avoid redundant downloads. The architecture supports identical residual processing from either the web interface or command-line evaluation scripts.

3.2 Generative noise model

The noise model is a 1D U-Net [Ronneberger et al., 2015] with four encoder/decoder stages (16 base channels, kernel size 9, GELU activations, batch normalization). It maps whitened strain S_{raw} to a predicted noise waveform $\hat{N}$, yielding

$$R = S_{\mathrm{raw}} - \hat{N}. \tag{2}$$

Training uses cached real H1/L1 background windows from GWOSC. A residual loss term encourages injected short bursts to survive subtraction rather than being absorbed into $\hat{N}$. A hold-out acceptance criterion requires mean normalized recovery error $\mathrm{RMS}(R - s)/\mathrm{RMS}(s) < 0.15$ on injections with known ground-truth signal s. The checkpoint used in this study is frozen as `2026-07-paper-v1` (SHA256 prefix `55ce7637...`).

3.3 Excess-power statistic

For a residual or raw series $x(t)$, Keno computes a short-time Fourier transform (Hann window, $n_{\mathrm{perseg}} = 256$, 75% overlap), forms bin power $P(f,t)$, estimates the noise floor as $\mathrm{median}(P)$, and defines

$$\rho_{\mathrm{EP}}(x) = \max_{f,t} \frac{P(f,t)}{\mathrm{median}(P)}. \tag{3}$$

This quantity measures how much the brightest spectrogram cell exceeds typical background; no template correlation is required.

3.4 Single-detector residual gate

The residual threshold θ_{IFO} is calibrated on noise-only windows at the target FAR. Because imperfect subtraction can produce occasional large artifacts, the upper 1% of noise-only residual scores is trimmed before computing the $(1 - \mathrm{FAR})$ percentile. In the frozen configuration, $\theta_{\mathrm{IFO}} \approx 4004.6$ at 1% FAR. Dual-detector production detection uses the coherent threshold below; this gate serves primarily for single-detector diagnostics.

3.5 Coherent dual-detector lag scan

Given Hanford and Livingston residuals R_{H} and R_{L}, Keno scans time shifts with $|\ell|/f_s \leq 10\,\mathrm{ms}$ (comparable to the inter-site light-travel time) and relative sign flips $p \in \{+1, -1\}$:

$$C_{\ell,p} = \frac{R_{\mathrm{H}} + p\,\mathcal{S}_\ell(R_{\mathrm{L}})}{\sqrt{2}}, \tag{4}$$

where $\mathcal{S}_\ell$ denotes zero-padded shifting. The coherent statistic is

$$\rho_{\mathrm{coh}} = \max_{\ell,p} \rho_{\mathrm{EP}}(C_{\ell,p}), \tag{5}$$

with best lag $\tau^\star = \ell^\star/f_s$. Astrophysical signals are expected to align within a few milliseconds across detectors; most glitches do not.

3.6 Envelope-consistency veto

Envelope peak times t_{H} and t_{L} are estimated from a smoothed energy envelope (8 ms boxcar), excluding the first and last 64 ms to limit edge effects. Define

$$\Delta t_{\mathrm{env}} = (t_{\mathrm{L}} - t_{\mathrm{H}}) \times 10^3 \quad (\mathrm{ms}). \tag{6}$$

When $|\Delta t_{\mathrm{env}}| > 50\,\mathrm{ms}$ and the coherent score remains high, the candidate is classified as likely single-detector contamination, as observed for the Livingston glitch near GW170817.

3.7 Production detection rule

A dual-detector production trigger is declared when

$$\rho_{\mathrm{coh}} \geq \theta_{\mathrm{coh}}, \tag{7}$$

$$|\tau^{\star}| \leq 10\,\mathrm{ms}, \tag{8}$$

$$|\Delta t_{\mathrm{env}}| \leq 50\,\mathrm{ms}. \tag{9}$$

The coherent threshold θ_{coh} is calibrated on dual-detector noise trials that satisfy the envelope gate. From 1500 noise trials (23 envelope-gated), the 99th percentile of gated coherent EP yields $\theta_{\mathrm{coh}} \approx 173.1$, corresponding to empirical joint FAR 0.07% (1/1500). After gating, coherent noise scores lie well below the single-detector residual threshold.

3.8 Baselines

All baselines use the same whitened windows and the same target FAR on noise-only trials:

- **Oracle matched filter** — correlation against the true injected waveform (injection studies only; upper bound).
- **Mismatched matched filter** — correlation against a fixed sine-Gaussian template that does not match the injected family.
- **Raw excess power** — $\rho_{\mathrm{EP}}(S_{\mathrm{raw}})$ without generative subtraction.
- **BBH-trained ResNet (control)** — a compact 1D ResNet trained on merger-like chirps versus noise in the same cache, scored as $P(\text{signal})$ on the central 1 s. This local morphology-specific control follows the design of AresGW [Nousi et al., 2023] but is not an external published checkpoint.
- **Keno residual EP** — $\rho_{\mathrm{EP}}(R)$.
- **Keno overlap** — cosine similarity between residual and injected signal (evaluation only; requires ground truth).

4 Experiments

Injection campaigns embed sine-Gaussian, ringdown, and white-noise bursts into cached real LIGO background at equal FAR. Additional studies include a glitch stress test on public O3 Gravity Spy labels [Gravity Spy et al., 2021], dual-detector noise coincidence FAR measurement, and follow-up at published GWTC/cWB GPS times from open catalogs. The catalog analysis is a consistency check, not a discovery claim.

5 Results

5.1 Unknown-morphology efficiency

Figure 2 shows Keno residual EP near 100% efficiency from SNR 2–12 at 1% FAR (freeze campaign; Wilson 95% intervals). On the same synthetic injections, raw excess power without subtraction also saturates at this FAR, so the plot does not by itself establish a residual advantage over raw excess power. Mismatched matched filtering remains near ∼20% and does not exceed 50%. The BBH-trained ResNet control, evaluated on these non-BBH bursts, plateaus below saturation, as expected for a morphology-specific model applied off-template. The principal residual-versus-raw contrast appears in the catalog follow-up below (e.g. GW150914 residual-only trigger).

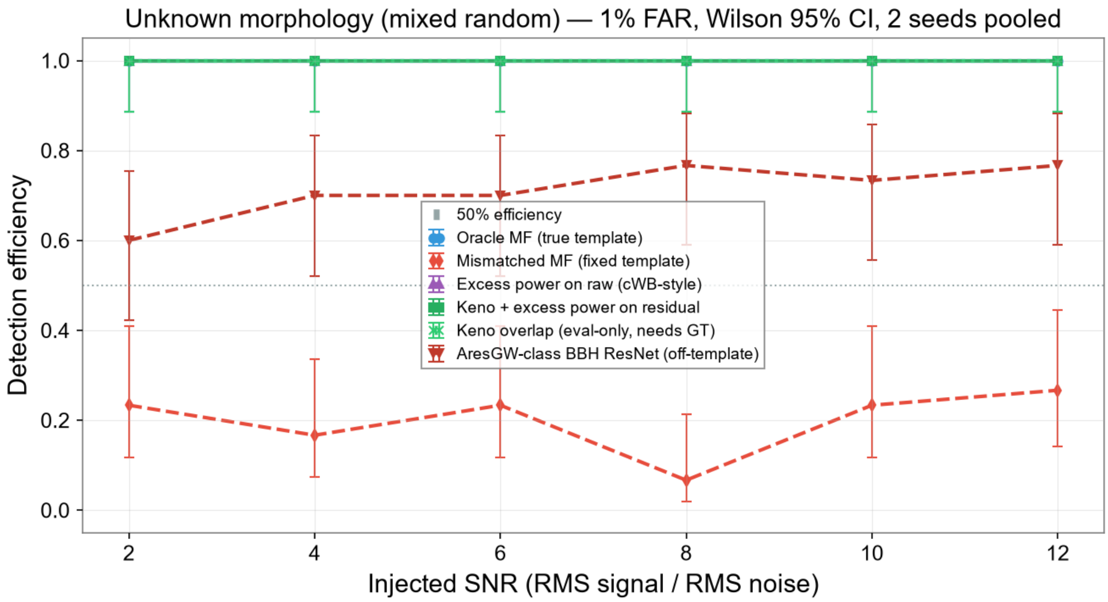

Figure 2: Detection efficiency versus injected SNR at 1% FAR for unknown morphologies. Keno residual EP and raw excess power both saturate on these injections; mismatched matched filtering stays low; the BBH-trained ResNet control is evaluated off-template.

5.2 Coherent calibration

Envelope gating removes most dual-detector noise outliers. With 1500 noise trials (23 envelope-gated), the coherent EP threshold is 173.1 at empirical joint FAR 0.07%, distinct from the single-detector residual threshold ≈ 4005.

5.3 Gravity Spy glitch taxonomy

Single-detector subtraction alone does not reliably suppress glitches. On a curated O3a Gravity Spy subset of 30 triggers spanning six labels, residual excess power still exceeded threshold for 23 cases (76.7%), whereas the single-detector suppression rate (raw trigger absent after subtraction) was 0/30. Injected unknown-morphology controls on the same background were preserved at 40/40 (residual trigger and overlap ≥ 0.75).

Residual survival by Gravity Spy class ($n = 5$ per label in this freeze):

- **Scattered Light** — lowest survival (20%). Its longer, noise-like morphology is partially absorbed into $\hat{N}$.
- **Blip** — moderate survival (60%). Short chirp-like transients are partially suppressed but often remain loud in R.
- **Tomte** — high survival (80%).
- **Koi Fish**, **Extremely Loud**, **Whistle** — 100% survival in this sample. These burst-like classes are retained in R similarly to astrophysical bursts.

Accordingly, production glitch mitigation relies on dual-detector coincidence and the envelope-timing veto rather than on single-interferometer subtraction to remove Gravity Spy morphologies.

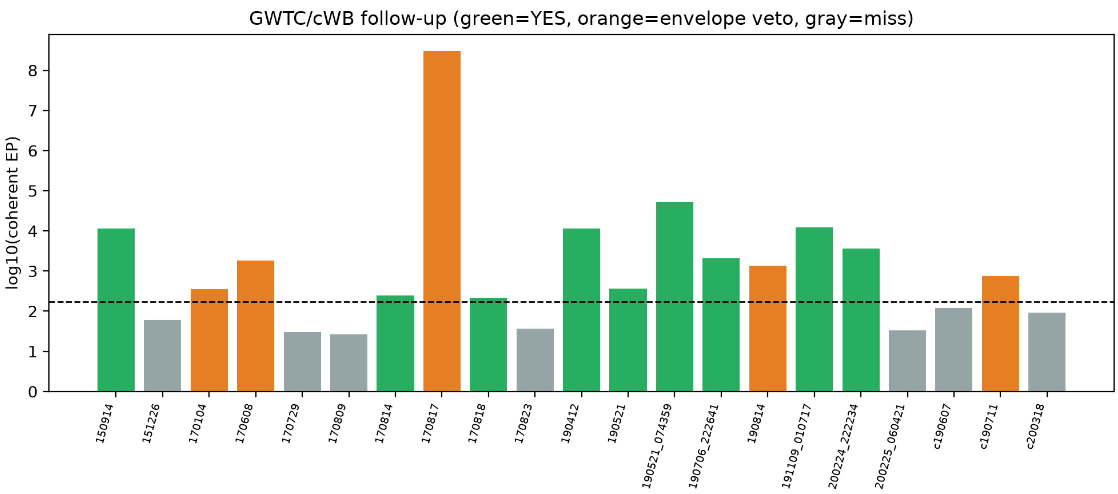

Figure 3: Coherent EP by event. Green: production detection; orange: envelope veto; gray: miss.

5.4 Catalog GPS follow-up

Production coincidence recovers 9 of 18 dual-detector catalog events (Figure 3). GW150914 is recovered with Independent=no and Production=yes: Hanford residual EP clears the single-detector gate while Livingston does not, yet the coherent lag scan still passes the dual-detector coherent threshold with aligned timing (EP $\approx 1.14\times10^4$, lag $+8.5$ ms, envelope peak dt -27.6 ms). That asymmetry is expected and illustrates a case where the residual path triggers while raw excess power does not. GW170817 is rejected by the envelope veto despite high coherent EP, owing to the Livingston glitch. Contaminated recoveries are excluded from the detection count. Three cWB-only candidates remain 0/3.

6 Discussion and limitations

Keno provides a reproducible software framework for unmodeled residual search. Morphology-specific BBH classifiers and compact-binary matched-filter pipelines [Nousi et al., 2023, Koloniari et al., 2025] remain appropriate when the target signal family is known.

Notable limitations of this prototype include:

- on the freeze injection campaign at 1% FAR, raw excess power and Keno residual EP both saturate, so synthetic efficiency alone neither proves nor fully characterizes a residual-versus-raw advantage;
- limited single-detector glitch rejection, particularly for burst-like Gravity Spy classes (Koi Fish, Whistle, Extremely Loud); coincidence requirements and the envelope veto provide the primary mitigation;
- residual over-subtraction if training coverage is insufficient, partially addressed by the recovery-error acceptance criterion;
- incomplete recovery of some catalog candidates in this configuration;
- network-bound latency on first-time GWOSC downloads (1–3 minutes), although cached windows are returned quickly;
- development by a single author without collaboration-scale detector characterization or production operations support.

7 Reproducibility

Source code, freeze artifacts, and reproduction scripts are public at `https://github.com/PanosM397/Keno`. Frozen artifacts live under `docs/freeze/current/` with checkpoint SHA256 prefix `55ce7637...` (label `2026-07-paper-v1`). Results may be reproduced with:

```
cd ml-engine
python -m app.evaluation.aresgw_class --train
python -m app.prove --freeze --freeze-label 2026-07-paper-v1
```

Interactive inspection uses the Angular frontend with the Express backend proxying the FastAPI ML engine; see repository READMEs for local setup.

8 Conclusion

Generative noise subtraction followed by template-free residual search can recover unknown-morphology bursts at controlled false-alarm rate in a reproducible research prototype. Keno implements this workflow as deployable software—inference API, orchestration layer, and synchronized residual visualization—intended for audit and extension alongside existing BBH classifiers, rather than as a finished gravitational-wave search pipeline.

Data availability

Code and the freeze bundle are available at `https://github.com/PanosM397/Keno` (label `2026-07-paper-v1`; checkpoint SHA256 prefix `55ce7637...`). A preprint archive of the manuscript, LaTeX source, and freeze metadata is deposited on Zenodo: `https://doi.org/10.5281/zenodo.21433068`. Large evaluation CSVs are listed with cryptographic hashes in the freeze manifest under `docs/freeze/current/`

References

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